

MUHAMMAD ZAIN FARRUKH

Artificial Intelligence & Machine Learning Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with proven experience designing and deploying end-to-end intelligent systems across computer vision, NLP, predictive ML, and LLM-based automation. Built a production-grade YOLOv8 PPE detection system processing 50 live RTSP camera feeds at 40 FPS, deployed at construction sites in Dammam (KSA) and Bahrain. Developed HillSafe AI, a full-stack disaster early warning platform combining ML risk prediction, a Django REST backend, and a Flutter mobile app with real-time geospatial alerts. Engineered an asynchronous Python Discord bot that monitors Upwork's internal GraphQL API for real-time job discovery, bypassing Cloudflare anti-bot protections with curl_cffi and automated session token refreshes. Experienced in NLP pipelines (fake news detection), medical ML classifiers (diabetes prediction, 80% accuracy), LLM-powered chatbots, and n8n business workflow automation. Proficient in Python, PyTorch, Django, FastAPI, Flutter, PostgreSQL, and AWS. Certified in Deep Learning (DeepLearning.AI) and AWS Cloud Foundations. Completing final semester of BS Computer Science at University of Central Punjab.

WORK EXPERIENCE

Automation Engineer Intern · Bitbash

Apr 2026 · Jun 2026

Lahore, Pakistan

- Developed an asynchronous Python Discord bot monitoring Upwork's internal GraphQL API for keyword-specific jobs in real time, using asyncio, curl_cffi, and SeleniumBase to bypass Cloudflare anti-bot protections.
- Built Crypto Sentry, a full-stack cryptocurrency monitoring platform (Next.js, PostgreSQL, Prisma, Auth.js) with live CoinGecko ingestion, watchlist management, flash-movement alerts, and dark terminal-style dashboard.
- Developed Spotify Automation Bot · end-to-end Android automation via Java Accessibility Service, FastAPI WebSocket backend, and React dashboard for remote multi-device Spotify control.
- Implemented SQLite-based persistence and dynamically routed formatted job alerts to dedicated Discord channels.

AI/ML Engineer · Freelance (PPE Detection System)

2026

Remote | github.com/mzain6/PPE-Detection

- Engineered a production-grade PPE detection system using YOLOv8, handling 50 concurrent RTSP streams at 40 FPS on GPU infrastructure · deployed at active construction sites in Dammam, KSA and Bahrain.
- Trained custom YOLOv5/YOLOv8 models on annotated safety datasets using transfer learning; benchmarked variants for speed-accuracy trade-offs and optimised GPU inference for real-time multi-stream processing.

React Developer · DevSaga

Sep 2023 · Mar 2024

Lahore, Pakistan

- Built reusable UI component libraries in React.js (ES6+) integrated with REST APIs, reducing frontend development time.
- Optimised component lifecycle and state management, improving rendering performance and reducing UI latency.
- Participated in full agile sprint cycles · planning, development, code review, and deployment · using Git.

PROJECTS

Crypto Sentry · Cryptocurrency Monitoring Dashboard

Full-Stack Web App | Bitbash, 2026 | github.com/mzain6/Crypto-Sentry

- Built a full-stack crypto monitoring platform with Next.js, PostgreSQL, Prisma ORM, and Auth.js supporting email/password and Google OAuth with JWT sessions and Resend-powered forgot-password flows.
- Implemented live CoinGecko ingestion (30s intervals), watchlist management, flash-movement alert evaluation, and per-user configurable threshold settings persisted in PostgreSQL.
- Developed dark terminal-style dashboard with real-time portfolio overview, sentry analytics, system alerts panel, and guided tutorial; deployed to Vercel with Docker-compose support.

Tech: Next.js, TypeScript, PostgreSQL, Prisma, Auth.js, CoinGecko API, Framer Motion, Resend

Spotify Automation Bot

Android Automation | Bitbash, 2026 | github.com/mzain6/Spotify-Automation

- Built an end-to-end Spotify automation system controlling multiple Android devices from a React dashboard with real-time WebSocket command dispatch and independent per-device result tracking.
- Developed an Android APK using Java Accessibility Service for UI-tree-based Spotify automation with persistent device identity binding, host normalisation, and ADB transport management.
- Engineered a FastAPI backend with SQLite persistence, per-device command routing, auto-APK relaunch, and JWT-protected dashboard authentication.

Tech: Python, FastAPI, WebSocket, SQLite, Java, Android Accessibility Service, React, ADB

Upwork Job Monitor Bot

Discord Bot | Bitbash, 2026 | github.com/mzain6/Discord-Bot

- Developed an asynchronous Python Discord bot monitoring Upwork's internal GraphQL API for keyword-specific jobs in real time, bypassing Cloudflare anti-bot protections with curl_cffi and SeleniumBase.
- Engineered a non-blocking async architecture with automated zero-downtime session token refreshes.
- Implemented an SQLite database for 100% duplicate-free tracking; routed formatted job alerts into dedicated Discord channels via threaded embeds.

Tech: Python, asyncio, curl_cffi, SeleniumBase, SQLite, Discord API

HillSafe AI · Disaster Early Warning System

Full-Stack AI System | Final Year Project, 2025-2026 | github.com/syedhasht/HillSafe-Ai

- Led the AI/ML component of a collaborative final year project; designed and trained ML models for environmental hazard prediction (landslides, floods) using historical weather and terrain datasets.
- Built and integrated a Django REST Framework backend managing alerts, regions, risk predictions, and real-time sensor/weather API ingestion.
- Collaborated on a Flutter mobile app with interactive geospatial risk maps (flutter_map) and Firebase push notification alerts for emergency warnings.

Tech: Django, Python, Flutter, Firebase, PostgreSQL, Machine Learning, REST APIs

Diabetes Prediction System

ML Classification | 2025 | github.com/mzain6/sugar-sense-react

- Built an SVM-based classifier achieving 80% accuracy for early diabetes detection from medical indicators.
- Applied data cleaning, normalisation, feature scaling, and hyperparameter tuning; evaluated with precision, recall, and confusion matrix analysis.

Tech: Python, Scikit-learn, Pandas, NumPy

Fake News Detection System

NLP | 2025 | github.com/mzain6/Fake-News-detection-using-NLP

- Designed an NLP pipeline using TF-IDF vectorisation and trained Logistic Regression and Random Forest classifiers for fake news classification.
- Applied text preprocessing (tokenisation, stop-word removal, stemming) within a modular pipeline.

Tech: Python, Scikit-learn, NLP, TF-IDF

Tutor LLM · RAG Application

RAG Application | 2025 | github.com/mzain6/Tutor-LLM

- Built a full-stack RAG application enabling students to upload a course syllabus (PDF/TXT) and ask contextual questions grounded in the document.
- Integrated Ollama (llama3.2) as a local LLM with nomic-embed-text embeddings and ChromaDB for vector storage.
- Developed a FastAPI backend and React/Vite frontend for a seamless end-to-end experience.

Tech: Python, FastAPI, React, Vite, Ollama, ChromaDB, LLM, RAG

n8n Business Workflow Automation

AI Automation | 2025

- Developed AI-powered automation pipelines for lead capture, appointment scheduling, CRM updates, and customer support, significantly reducing manual workload.
- Integrated chatbots with business tools via n8n nodes, webhooks, and email triggers for end-to-end automation.

Tech: n8n, Webhooks, JSON, REST APIs

CERTIFICATIONS

Neural Networks and Deep Learning · DeepLearning.AI / Coursera

coursera.org/share/c4830b495925a7d76cc868ad99ba6b80

AWS Academy Graduate · Cloud Foundations · Amazon Web Services

credly.com/badges/c2a82fbc-dc6e-4bad-a1e5-9407db9c3f65/public_url

SKILLS

AI / ML	YOLOv5/v8, Computer Vision, OpenCV, Scikit-learn, PyTorch, SVM, Random Forest, LSTM, Transformers, NLP, TF-IDF, Diffusion Models
Data & ML Ops	Pandas, NumPy, Matplotlib, Feature Engineering, Data Preprocessing, Model Evaluation, Hyperparameter Tuning, CUDA
Backend	FastAPI, Django REST Framework, Flask, REST API Design, PostgreSQL, MySQL, Firebase
Automation & LLMs	n8n, Open-source GPT Models, LLM Integration, Webhooks, JSON Pipelines
Cloud & Tools	AWS (Cloud Foundations), Git, Docker (basic), VS Code, UML
Languages	Python, C++, Dart, Java, C
Frontend / Mobile	React.js, Next.js, HTML, CSS

EDUCATION

Bachelor of Science in Computer Science

University of Central Punjab, Lahore | Oct 2022 – Present (Final Semester)